Fit Test Plate

Find the clearances that work on **your** printer, with **your** filament, in one print. Five mechanisms, each printed at three clearances. Put each pair together, feel which one is right, and use that number in everything you design.



A real print of this plate, straight off a Bambu Lab P1S: 33 parts, no supports.

Five mechanisms

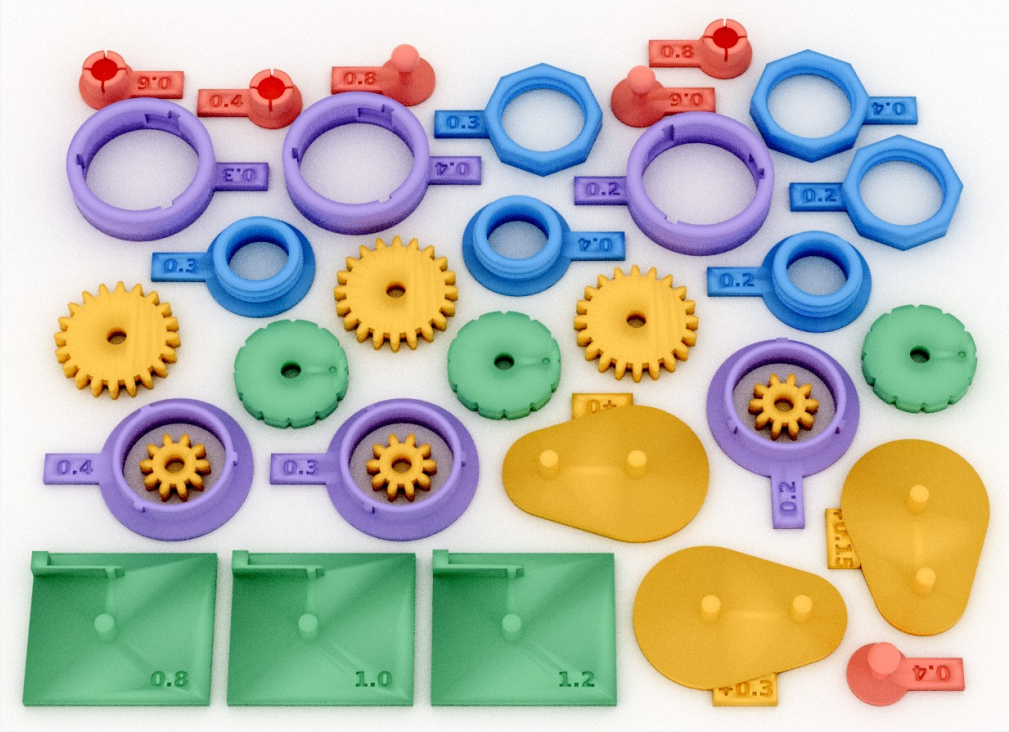
- **Thread** 24 mm screw and nut, pitch 3
- **Ball joint** 8 mm ball, snap-in socket
- **Gears** module 1.5, 20 and 10 teeth
- **Detent** 12-click wheel on a spring arm
- **Bayonet** twist-lock cap and ring

Print it like this

- PLA you'll use for real parts. One colour is fine.
- 0.20 mm layers, 0.4 mm nozzle, default 2 walls.
- **Supports off.** None of the 33 parts needs any.
- **Don't scale it.** The clearances are real millimetres.
- Sliced estimate: about 3 h 40 min and 93 g. 16 mm tall.

What's on the plate

Each colour below is one mechanism: three pairs, one at each clearance. In every pair **one part changes and the other is the same in all three**. The part that changes carries its clearance, engraved on a little tab or in its base. Some fixed parts carry a matching label too, just to keep pairs together.



Mechanism	The part that changes	Its labels	What the number means
Thread	the octagonal nut	**0.2 · 0.3 · 0.4**	gap all round the thread
Ball joint	the split socket	**0.4 · 0.6 · 0.8**	mouth this much under the ball
Gears	the pear-shaped base	**+0 · +0.15 · +0.3**	extra gap between gear centres
Detent	the square base	**0.8 · 1.0 · 1.2**	thickness of the spring arm
Bayonet	the cap with lugs	**0.2 · 0.3 · 0.4**	gap round the lugs

Before you start

1. Take everything off the plate and sort it into the five mechanisms. Three of the small 10-tooth gears were packed inside the bayonet rings to save space: pop them out.
2. Line up each mechanism's three pairs by the labels on the parts that change.
3. **Don't sand, trim or force anything.** The point is how they come off your printer. If something only sticks on a squashed first layer (elephant's foot), try the part the other way up and note it.
4. Test the three clearances of one mechanism back to back, so you compare like with like. Tick the results sheet on page 8 as you go.

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Thread

Tests how much gap a printed thread needs to turn by hand.

Labels on the **nut**: 0.2 · 0.3 · 0.4, the gap all round the thread in mm. The screws are labelled too, to keep pairs together, but all three are the same.

1 Line up
the nut over the post



2 Screw it down
clockwise, by fingertip



Steps

1. Hold the screw by its flange, post up. Either face of the nut can go down: both ends have a lead-in.
2. Set the nut on the post and turn it **clockwise** (an ordinary right-hand thread). It should catch within half a turn.
3. Keep turning with fingertips only, all the way down to the flange: just over two turns.
4. Back it off a turn and rock the nut side to side. Note how much it tilts or clunks.
5. Run it on and off five times. Does it get smoother or rougher? Any shavings?

How to judge it

Too tight
Needs a hard grip, squeaks, or stops partway.

Just right
Turns all the way by fingertip, no grinding, only a little rock.

Too loose
Tilts visibly, or could jump a thread if you pulled it.

If two values both pass, the tighter one that still turns by fingertip is usually the better pick: it lines up more accurately when it closes.

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Ball joint

Tests how hard a ball joint should snap in to hold a pose.

Labels on the **socket**: 0.4 · 0.6 · 0.8, how much narrower its mouth is than the 8 mm ball. The bigger the number, the harder the snap. The balls are labelled too, but all three are the same.

1 Line up
ball turned over, above the cup



2 Snap it in
straight down, by thumb



Steps

1. Stand the socket on the table, cup up. Turn the ball part upside down so the ball points into the cup.
2. Press straight down with your thumb until it snaps in. If a hard thumb press won't do it, stop: that's too tight.
3. Hold the socket's disc and tilt the stem to about 30°. Let go. Does it stay? Give it a gentle shake.
4. Swivel the stem round in a circle. Smooth, or does it catch?
5. Pull the two straight apart. Then look at the cup's four fingers for white stress marks or a crack.

How to judge it

Too tight
Won't go in by thumb, or the fingers turn white.

Just right
A firm snap, holds 30° without drooping, pulls out with real effort.

Too loose
Droops from 30°, or falls out when shaken.

For posable figures, holding the angle matters more than how easily it goes in.

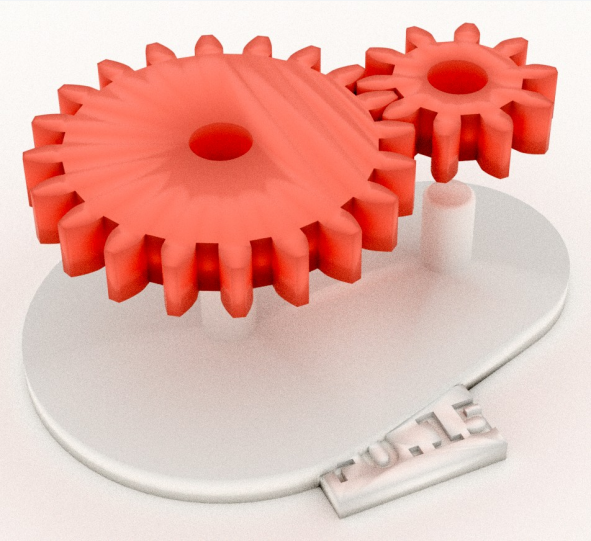
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Gears

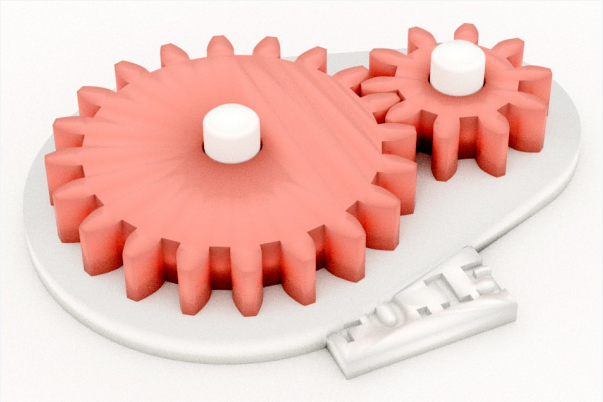
Tests how far apart two printed gears should sit to turn smoothly without skipping.

Labels on the **base**: +0 · +0.15 · +0.3, extra space between the gears' centres in mm. All the gears are the same.

1 Line up
both gears over their pins



2 Mesh and turn
teeth into teeth



Steps

1. Press a 20-tooth gear onto the pin at the wide, round end of a base. It should spin freely on its own.
2. Press a 10-tooth gear onto the other pin, turning it slightly so its teeth drop into the big gear's.
3. Turn the big gear slowly through one full turn (the small one makes two). Feel for a tight spot.
4. Hold the big gear still and wiggle the small one. A small tick of play is normal.
5. Drive it from the small gear while pushing it toward the big one. Do the teeth ever skip?

How to judge it

Too tight
Binds or drags somewhere in the turn.

Just right
One smooth turn, a little play, never skips.

Too loose
Clacks, rattles, or skips teeth under a push.

If a gear is stiff on its pin before it meshes with anything, that's the pin, not the mesh: note it separately.

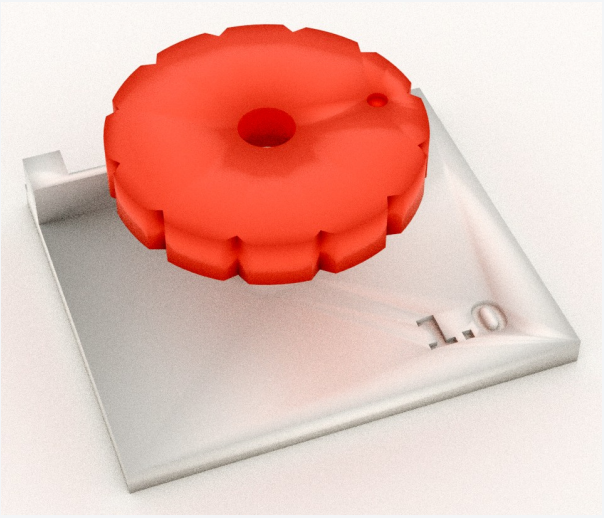
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Detent

Tests how thick a printed spring arm should be to give a crisp click that lasts.

Labels engraved in the **base**: 0.8 · 1.0 · 1.2, the thickness of the spring arm in mm. The three notched wheels are the same.

1 Line up
the wheel over the pin



2 Press down and turn
12 clicks a turn



Steps

1. Set a wheel over the pin, dot up. Turn it so a notch faces the arm's bump (or push the arm back with a fingernail) and press it flat.
2. Turn the wheel one full turn by its edge: 12 clicks, counted by the dot. It should never rest between two notches.
3. Note the feel: a crisp click, a mushy one, or too stiff for a fingertip.
4. Turn it three full turns each way. Watch where the arm meets its post for white marks or a crack.
5. With the wheel in a notch, tap the base on the table. The wheel shouldn't move.

How to judge it

Too tight
Hard to turn, or the arm turns white.

Just right
A crisp click on all 12, turns by fingertip, no whitening.

Too loose
A faint click, or the wheel drifts off a notch.

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Bayonet

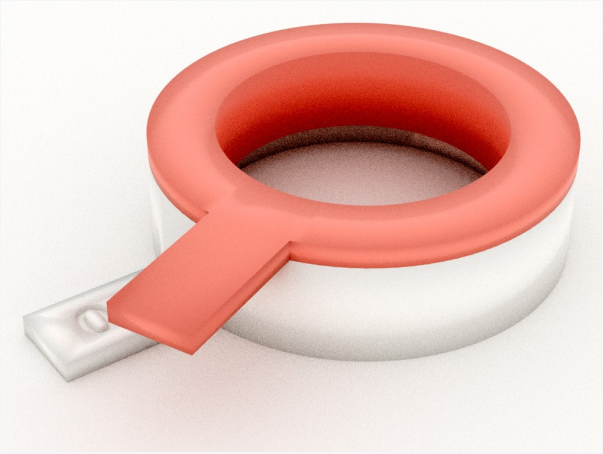
Tests how much room a twist-lock lid needs to drop in, turn and hold.

Labels on the **cap**: 0.2 · 0.3 · 0.4, the gap round its three lugs in mm. The rings are labelled too, but all three are the same.

1 Line up
lugs over the three openings



2 Push down, twist left
about 24°, to the stop



Steps

1. Turn the cap over, plug down. Look into the ring from above and find its three openings.
2. Line the cap's three lugs up with the openings and push it straight down until it stops.
3. Twist it **counterclockwise** (left, seen from above) to the stop. It's a short twist, about 24°.
4. Locked, try to lift the cap straight up. It shouldn't come out. Does it rattle or tilt?
5. Twist back to the openings, lift it out, and repeat five times. Then check the lugs for chips.

How to judge it

Too tight
Won't drop in, or the twist needs force.

Just right
Drops in, twists smoothly to a firm stop, won't lift, barely rattles.

Too loose
Rattles, tilts, or lifts while locked.

Results sheet

Print this page, or keep it on screen and mark it up. Tick one box per clearance, then circle your pick for each mechanism.

		Too tight	Just right	Too loose	Broke	Notes
Thread	0.2	☐	☐	☐	☐	
	0.3	☐	☐	☐	☐	
	0.4	☐	☐	☐	☐	
My pick: 0.2	0.3	0.4	none			
Ball joint	0.4	☐	☐	☐	☐	
	0.6	☐	☐	☐	☐	
	0.8	☐	☐	☐	☐	
My pick: 0.4	0.6	0.8	none			
Gears	+0	☐	☐	☐	☐	
	+0.15	☐	☐	☐	☐	
	+0.3	☐	☐	☐	☐	
My pick: +0	+0.15	+0.3	none			
Detent	0.8	☐	☐	☐	☐	
	1.0	☐	☐	☐	☐	
	1.2	☐	☐	☐	☐	
My pick: 0.8	1.0	1.2	none			
Bayonet	0.2	☐	☐	☐	☐	
	0.3	☐	☐	☐	☐	
	0.4	☐	☐	☐	☐	
My pick: 0.2	0.3	0.4	none			

Printer, filament and settings, so the numbers mean something later: _____

Using your numbers

Each pick is a clearance your printer really produces, not a guess from a forum. Use it the next time you design that kind of part, for the same filament and settings.

Thread

Grow the nut (or shrink the screw) by your pick, all round. This plate's thread is 24 mm across with a 3 mm pitch and a coarse 0.8 mm tooth; a finer thread may want a little more room.

Ball joint

Make the socket's mouth narrower than the ball by your pick. The cup needs slits to flex over the ball: these sockets have four, 0.8 mm wide, round an 8 mm ball.

Gears

Add your pick to the gears' theoretical centre distance (the sum of their pitch radii). Here that's 22.5 mm for module 1.5 with 20 and 10 teeth, so +0.15 means 22.65 mm.

Detent

Make a spring arm about your pick thick, in the same orientation. These arms print flat, 19 mm long, and flex 0.6 mm per click. An arm standing upright is weaker across its layers: if your pick whitened at all, use the next one down for upright arms.

Bayonet

Leave your pick as the gap round the lugs, in the slot and on the way in. These lugs are 1.4 mm deep in a 32 mm bore, and lock with a 24° counterclockwise twist.

If none of the three worked

Note which way they all failed: all too tight, or all too loose. Then step past the end of the range by the same step (for a thread that was too tight at 0.4, try 0.5). A problem in every mechanism at once usually means the printer itself: calibrate flow and pressure advance for the filament, check for elephant's foot, and print again.

Share your numbers in the comments. Say which printer, nozzle and filament. Every report helps the next person start closer to right.